

Consistency of Surrender Value Calculations under Paid-Up Policy States

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Overview

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- Valuation pipeline for surrender value
- Scenario:
 - premium payment stopped (paid-up state)
 - subsequent surrender event
- Observed issue:
 - output not aligned with expected economic behavior

Core Modeling Problem

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- Paid-up state implies:
 - only core contract remains economically active
 - riders effectively inactive
- Internal representation:
 - riders removed from structure
 - but still included in aggregated premium inputs
- Result:
 - mismatch between product structure and cashflow inputs

Source of Inconsistency

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- Valuation logic assumes:
 - premium inputs correspond to active components
- In paid-up scenario:
 - component structure and premium inputs diverge
- Consequence:
 - internal quantities derived from inconsistent basis
 - distortion of surrender value

Implementation Approach

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- Solution applied at input level of valuation
- Strategy:
 1. store original model state
 2. adjust inputs to reflect paid-up structure
 3. execute standard valuation
 4. restore original state
- Key idea:
 - align premium inputs with active components only

What Was Actually Solved

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- Not a numerical issue
- Not a formula issue
- Core problem:
 - inconsistency between product state and valuation inputs
- Resolution:
 - enforce state-consistent inputs prior to valuation

Outcome

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- Surrender value consistent with paid-up economic interpretation
- No changes to core valuation logic
- No side effects for:
 - active policies
 - other product configurations
- Minimal, localized intervention